Engineering reference: governed manufacturing reporting

A sanitized technical case study covering architecture, contracts, state, validation, failure handling, tradeoffs, and impact methodology.

Reading guide

QUESTION	SECTION
How is the platform used?	1. Operational contract
What is implemented?	2. Architecture and data model
Why is delivery reliable?	3. State, validation, and recovery
How is evidence tested?	4. Verification matrix
What else was delivered or prepared?	5. Additional initiatives
How are impact figures qualified?	6. Impact methodology

Public/private boundary

This reference uses synthetic data and sanitized topology. Tenant identifiers, employee data, production URLs, credentials, exact commands, permissions, and successor procedures remain in the private handoff.

1. Operational contract

The platform prepares decision-support evidence; people retain operational authority.

OUTPUT	PRIMARY USER	PLATFORM RESPONSIBILITY	HUMAN RESPONSIBILITY
Shift Draft	Supervisor	Collect, reconcile, organize, prepare spreadsheet	Review exceptions, add context, finalize
Daily Production	Manager	Combine production, molding, labor, quality, plan, downtime	Investigate constraints and prioritize follow-up
Weekly Production	Managers / teams	Create comparable period and trend views	Plan capacity and labor; prioritize improvement
Weekly Downtime	Managers / maintenance	Classify recurring and isolated downtime evidence	Set maintenance and coordination priorities

Decision boundary

Automation may prepare, classify, validate, refuse, and report state. It does not autonomously change staffing, production plans, maintenance priorities, or cross-department actions.

2. Architecture and data model

LAYER	COMPONENTS	RESPONSIBILITY
Operational inputs	Email, network files, Excel source reports	Preserve the evidence people already produce
Reporting core	Python ingestion/reconciliation, PostgreSQL	Normalize evidence and maintain source-linked serving state
Controlled outputs	Workbooks and SharePoint delivery	Expose prepared shift, daily, weekly, and downtime reports
Additional initiatives	Forms/QR, machine events, notifications, SPFx dashboard	Extend intake and machine-status visibility without changing decision authority

Eight source-linked reporting tables

The serving model separates hourly production, quality, labor, molding, machine notes/events, final reports, report state, and lineage/settlement evidence. Separation prevents an output-friendly aggregate from erasing source identity or missingness.

CONTRACT	WHY EXPLICIT
Report identity	Prevents one period/family from settling another
Material identity	Distinguishes governed fact changes from capture-only metadata
Source lineage	Connects each prepared value to qualified evidence
Semantic certification	Tests meaning rather than file appearance alone
Publication/readback state	Separates accepted write from verified remote content

3. State, validation, and recovery

CANDIDATE	VALIDATED	PUBLISHED	READ BACK	SETTLED

State advances only when the evidence required for that transition exists. Missing evidence remains missing; an observed zero remains zero. A successful request is not equivalent to a verified remote artifact.

Failure scenarios

SCENARIO	CLASSIFICATION	SAFE BEHAVIOR	AUTHORITY
Required source absent	pending_source	Retain expected output; do not publish	No write
Equivalent replay	exact_no_op	Conserve material identity and remote version	No write
Write accepted; response lost	readback_pending	GET exact item; never repeat PUT	GET only
Contract or lineage drift	contract_refusal	Stop interpretation/publication	No write
Eligible late source	bounded_recovery	Advance same output identity once after validation	One qualified write

Release and rollback

One-writer releases serialize ownership and prevent overlapping publication. Promotion is guarded by output-suppressed/current-data checks. Exact rollback restores the last known-good immutable revision rather than reconstructing a deployment from mutable state.

4. Verification matrix

LAYER	REPRESENTATIVE CHECK	FAILURE PREVENTED
Source facts	Conservation and reconciliation against qualified inputs	Dropped or duplicated evidence
Workbook package	Visible sheets, formulas, structure, stored values	Client-dependent or malformed output
Semantics	Missing-versus-zero and family-specific rules	Meaning drift
Normalized rendering	Stable content independent of packaging noise	False changes from metadata
Remote readback	Exact item, version, ETag, content/material hash	Accepted-write ambiguity
Replay	Immediate equivalent cycle makes zero second PUT	Duplicate versions
Release	Serialized handoff, readiness guard, exact rollback	Concurrent writers / unsafe promotion

Claim discipline

Validation coverage proves the tested invariants and scenarios. It is not presented as blanket accuracy. Public evidence is sanitized; production-specific receipts and exact successor tests remain private.

Selected tradeoffs

CHOICE	BENEFIT	COST
Fail closed	No silent advancement on ambiguity	Some reports settle later
One writer	Clear ownership and recovery semantics	Serialized promotion
Same-item advancement	Stable operational identity and links	Requires strict replay/material checks
Bounded repair	Recovers only evidence-supported cases	Some historical cases remain terminal holds

5. Additional initiatives

INITIATIVE	STATUS	PURPOSE	BOUNDARY
Machine-event visibility	Delivered extension	QR/Form intake, notifications, and SharePoint dashboard status	Visibility only; people investigate and decide
Reporting Assistant	Functional read-only extension	Answers bounded queries with source-grounded retrieval, citations, diagnostics, validation, and refusal	No write authority; broader model-backed/advisory work is future

Technical foundation

The assistant separates bounded retrieval, read-only report-doctor diagnostics, and governed-oracle validation because they have different authority and refusal behavior. Those functional paths reuse governed identity, lineage, state, and semantic contracts; they do not create a new writer.

Human control

Machine visibility and bounded retrieval can shorten the path to evidence. Supervisors and managers still interpret context and choose operational action.

6. Impact methodology

VALUE AREA	BASIS	RESULT	QUALIFICATION
Core reporting	Underlying company-specific assumptions are omitted from this public case study.	30 hours/week; 1,560 hours/year; about $52K	Modeled labor-capacity value, not booked savings
Assistant retrieval	Representative 20-30 minute search compared with a 1-2 minute cited response	About 90% faster	Estimated from the functional assistant; not a controlled production benchmark
Machine visibility	Underlying company-specific assumptions are omitted from this public case study.	Up to 227,500 production units; about $205K contribution margin	Additional-initiative upper bound, not realized production or margin

Limitations

The labor, assistant-retrieval, and machine-visibility figures represent separate value pools and are not additive. The models depend on representative time studies, event frequency, throughput, response improvement, role rates, and contribution margin. They support prioritization, not guarantees. The synthetic workbook separates demonstrated controls from documented impact assumptions.

Implementation references

The public repository contains sanitized architecture, state-machine documentation, tests, synthetic scenarios, release chronology, and claim boundaries. The private successor package contains exact dependencies, permissions, operating procedures, rollback, credentials transfer, and acceptance checks.